Genesmanship

p. 117* … *I have never been able to understand why they have been so neglected …

Hamilton's 1964 papers are neglected no longer. The history of their earlier neglect and subsequent recognition makes an interesting quantitative study in its own right, a case study in the incorporation of a 'meme' into the meme pool. I trace the progress of this meme in the notes to Chapter 11.

p. 117* … *I shall assume that we are talking about genes that are rare …

The device of assuming that we are talking about a gene that is rare in the population as a whole was a bit of a cheat, to make the measuring of relatedness easy to explain. One of Hamilton's main achievements was to show that his conclusions follow *regardless* of whether the gene concerned is rare or common. This turns out to be an aspect of the theory that people find difficult to understand.

The problem of measuring relatedness trips many of us up in the following way. Any two members of a species, whether they belong to the same family or not, usually share more than 90 per cent of their genes. What, then, are we talking about when we speak of the relatedness between brothers as, or between first cousins as? The answer is that brothers share of their genes *over and above* the 90 per cent (or whatever it is) that all individuals share in any case. There is a kind of baseline relatedness, shared by all members of a species; indeed, to a lesser extent, shared by members of other species. Altruism is expected towards individuals whose relatedness is higher than the baseline, whatever the baseline happens to be.

In the first edition, I evaded the problem by using the trick of talking about rare genes. This is correct as far as it goes, but it doesn't go far enough. Hamilton himself wrote of genes being 'identical by descent', but this presents difficulties of its own, as Alan Grafen has shown. Other writers did not even acknowledge that there was a problem, and simply spoke of absolute percentages of shared genes, which is a definite and positive error. Such careless talk did lead to serious misunderstandings. For instance a distinguished anthropologist, in the course of a bitter attack on 'sociobiology' published in 1978, tried to argue that if we took kin selection seriously we should expect all humans to be altruistic to one another, since all humans share more than 99 per cent of their genes. I have been given a brief reply to this error in my 'Twelve Misunderstandings of Kin Selection' (it rates as Misunderstanding Number 5). The other eleven misunderstandings are worth a look, too.

Alan Grafen gives what may be the definitive solution to the problem of measuring relatedness in his 'Geometric View of Relatedness', which I shall not attempt to expound here. And in another paper, 'Natural Selection, Kin Selection and Group Selection', Grafen clears up a further common and important problem, namely the widespread misuse of Hamilton's concept of 'inclusive fitness'. He also tells us the right and wrong way to calculate costs and benefits to genetic relatives.

p. 121* … *armadillos … it would be well worth somebody's while going out to South America to have a look.

No developments are reported on the armadillo front, but some spectacular new facts have come to light for another group of 'cloning' animals—aphids. It has long been known that aphids (e.g. greenfly) reproduce asexually as well as sexually. If you see a crowd of aphids on a plant, the chances are that they are all members of an identical female clone, while those on the next-door plant will be members of a different clone. Theoretically these conditions are ideal for the evolution of kin-selected altruism. No actual instances of aphid altruism were known, however,